

Graph Neural Networks for Electricity Load Forecasting: A Comparative Study of GCN and GAT Architectures on Florida County Data

Shambhu Marasini
Department of Electrical and
Computer Engineering
University of Central Florida
Orlando, FL, USA
sh118731@ucf.edu

Pablo Rodriguez
Department of Electrical and
Computer Engineering
University of Central Florida
Orlando, FL, USA
pa233796@ucf.edu

Abstract—Short-term electricity load forecasting is important for grid stability, resource scheduling, and market operations. Most forecasting models treat each location on its own, which means they ignore the fact that nearby regions usually see similar demand patterns. In this paper, we build a graph of Florida’s 67 counties where edges connect counties that are geographically close, and use this graph to improve hourly load forecasting. We compare three models: a Feed-Forward Neural Network (FF-NN) that does not use any geographic information, a Graph Convolutional Network (GCN), and a Graph Attention Network (GAT). All models were trained on 2021–2022 load data and tested on 2023 data using MAPE and RMSE. The GCN graph model outperformed the baseline feed-forward model by about 33%. We made our code and framework available at: <https://github.com/pa233796/FLpowerGNN>

Index Terms—Graph Neural Networks, GCN, GAT, Electricity Load Forecasting, Spatial Modeling, Florida Power Grid.

I. INTRODUCTION

A. Background and Motivation

Electricity is different from most other commodities because it has to be produced and consumed at almost the same time. Large-scale storage is still not really practical. This means power companies need to predict how much electricity will be needed hours ahead in order to plan generation.

This problem has gotten harder over the past decade. Electric vehicles and rooftop solar are changing demand in ways older models did not account for. Climate change is also producing hotter summers and more extreme weather events, which can push demand to unexpected highs in a short amount of time.

One thing that often gets overlooked is the spatial correlation between neighboring regions. On a hot summer afternoon in one Florida county, the next county over is almost certainly just as hot and pulling just as much power for air conditioning. If a model can pass that kind of information between neighbors, it should make better predictions than one that treats each county on its own.

B. Problem Statement

This project asks a practical question: does modeling Florida counties as a connected graph, instead of as 67 separate time series, give better hour-ahead electricity load forecasts? And between the two graph models we test, does the GAT’s learnable attention give a real advantage over the simpler averaging done by the GCN?

C. Our Contributions

- We adapt the GNN forecasting framework from Campaign *et al.* [1] to a U.S. county-level setting for the first time.
- We build a geographic graph of Florida’s 67 counties using a Gaussian distance kernel and apply it to real hourly load data from the NREL OEDI dataset [2].
- We do a clean three-way comparison of FF-NN, GCN, and GAT under the same hyperparameters and the same 2021–2022 training / 2023 test split.

II. RELATED WORK

A. Classical Forecasting Methods

For many years, Generalized Additive Models (GAMs) were the standard tool for electricity load forecasting [3]. They break demand down into smooth components for time of day, day of week, and temperature, and then add them up. They are easy to interpret and work reasonably well under normal conditions, but they need manual feature design and treat each location on its own. With power distribution networks becoming more decentralized, recent literature suggests that approaches capable of modeling decentralized networks, such as graph neural networks, may be a better fit for this task [4].

B. Deep Learning for Load Forecasting

Recurrent Neural Networks, especially Long Short-Term Memory (LSTM) models, have shown strong results because they can capture temporal correlations in load series [5].

However, these models still process each node on its own and do not capture interactions between nodes, which limits their ability to use spatial structure.

C. Graph Neural Networks for Energy Systems

GNNs were originally developed for things like social networks and chemistry, but they apply naturally to any domain that has a relational structure. Traffic forecasting was one of the early success stories, and cars moving along a road network is similar in structure to electricity flowing along a grid [6].

Campagne *et al.* [1] published one of the more comprehensive benchmarks of GNNs for electricity load forecasting. They tested GCN, GraphSAGE, APPNP, and GAT on French regional data and UK residential smart-meter data. They found that graph models consistently performed better than standard baselines. Our work follows their methodology closely and applies it to the counties of the U.S. state of Florida.

III. METHODOLOGY

A. Building the Graph

Each of the 67 Florida counties is one node in our graph. Two counties get an edge between them if they are geographically close. To measure closeness, we use a Gaussian kernel applied to the geodesic distance:

$$d(u, v) = \exp\left(-\frac{\text{dist}(u, v)^2}{\sigma_b^2}\right), \quad (1)$$

where $\text{dist}(u, v)$ is the straight-line distance between county centroids and σ_b is set to the median of all pairwise distances (about 164.18 miles for Florida). This gives a weight close to 1 for nearby counties and close to 0 for distant ones.

To keep the graph sparse, we drop any edge whose weight is below a threshold $\lambda = 0.2$:

$$W_\lambda = d(u, v) \cdot \mathbf{1}[d(u, v) \geq \lambda]. \quad (2)$$

The result is a connected but sparse graph where each county is linked to a few nearby neighbors, following the approach used in earlier studies.

B. Features and Targets

The input for each node at time t is its own load from the previous 24 hours, stored as a vector $\mathbf{x}_{v,t} \in \mathbb{R}^{24}$. The target $y_{v,t}$ is the load at the next hour. So this is a sequence-to-one setup where the model uses the previous day's pattern to predict the next step. All load values are normalized to $[0, 1]$ using `scikit-learn`'s `MinMaxScaler`.

C. Train and Test Split

We train on hourly demand data from January 2021 to December 2022 and test on data from 2023. The two windows do not overlap, so the evaluation reflects real out-of-sample generalization.

D. Evaluation Metrics

We report two metrics, matching the evaluation in Campagne *et al.* [1]. Mean Absolute Percentage Error (MAPE) gives the error as a percentage of the true value, which makes it easy to interpret:

$$\text{MAPE} = \frac{1}{T} \sum_{t=1}^T \left| \frac{\hat{y}_t - y_t}{y_t} \right| \times 100\%. \quad (3)$$

Root Mean Square Error (RMSE) is in physical units (megawatts) and penalizes large errors more than small ones:

$$\text{RMSE} = \sqrt{\frac{1}{T} \sum_{t=1}^T (\hat{y}_t - y_t)^2}. \quad (4)$$

Both metrics are computed on the aggregated statewide load, which is the sum of predictions across all 67 counties. This combination is commonly used in prior work and gives a reasonable picture of forecasting accuracy and stability.

IV. MODEL ARCHITECTURES

A. How GNNs Work

The basic idea behind a GNN is simple: every node starts with its own feature vector. At each layer, it gathers information from its neighbors, combines that with its current state, and updates itself. After a few layers, each node has been informed by a wider neighborhood. The general update rule is:

$$\mathbf{h}_v^{(\ell+1)} = \text{UPDATE}\left(\mathbf{h}_v^{(\ell)}, \text{AGG}\left\{\mathbf{h}_u^{(\ell)} : u \in \mathcal{N}(v)\right\}\right), \quad (5)$$

where $\mathbf{h}_v^{(0)} = \mathbf{x}_v$. The difference between GCN and GAT is in how they handle the AGG step.

B. Baseline: Feed-Forward Neural Network (FF-NN)

The FF-NN takes each county's 24-hour history and passes it through two fully connected layers, each with 32 hidden units, with a ReLU activation between them, before producing a single load prediction. It does not see the graph at all. If the graph models cannot beat this, then adding graph structure is not worth the extra effort.

C. Graph Convolutional Network (GCN)

The GCN takes the graph and features described in Sections III-A and III-B and runs them through two convolution layers with 32 hidden channels each and a ReLU activation. A dropout layer with $p = 0.2$ is placed between the convolution layers. The output of the second convolution layer is flattened and passed to a single linear fully-connected layer for the final regression task. The network uses the graph convolutional operator from [7].

D. Graph Attention Network (GAT)

The GAT is otherwise the same as the GCN, except that it uses the graph attentional operator from [8].

V. IMPLEMENTATION

A. Dataset

The power load data comes from the NREL OEDI hourly county-level demand dataset [2], which covers the entire contiguous United States from 2016 through 2023. We pulled out Florida data from 2021 to 2023. Geographic distances between counties for the graph come from the 2010 census [9].

B. Graph Construction Results

Before building the graph, we checked that all 67 Florida counties are present in both the load time-series and the distance datasets. This gave us 4,422 directed county pairs with no missing entries. After applying the Gaussian kernel (Eq. 1) with $\sigma = 164.18$ miles and thresholding at $\lambda = 0.2$ (Eq. 2), the edge count drops from 4,422 to 1,391, removing about 31% of the weakest long-distance connections.

Figure 1 shows the raw fully connected network. Every county pair has an edge, which creates a dense mesh that would dilute local demand signals during GNN message passing.

Figure 2 shows the graph after applying $\lambda = 0.2$. The uniform-edge view (Fig. 2) makes it easier to see which county pairs remain connected after pruning.

1. Before Thresholding (Fully Connected)

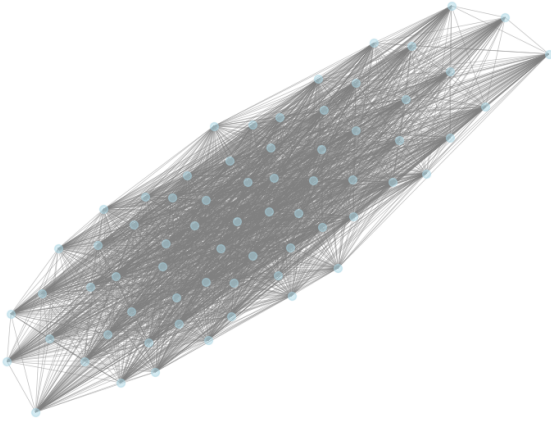


Fig. 1. Florida county graph **before** thresholding. 2211 edges in total.

C. Software Stack

All code is written in Python, organized into Jupyter notebooks, and runs on workstations with GPU acceleration. We use PyTorch for model training and PyTorch Geometric [10] for the graph convolution layers. pandas and numpy handle data loading and preprocessing, and matplotlib produces the plots, and certain utility functions from scikit-learn are also used. The code notebooks and required files to run the workflow are available on GitHub: <https://github.com/pa233796/FLpowerGNN>

2. After Thresholding ($\lambda=0.2$) - Uniform Edges

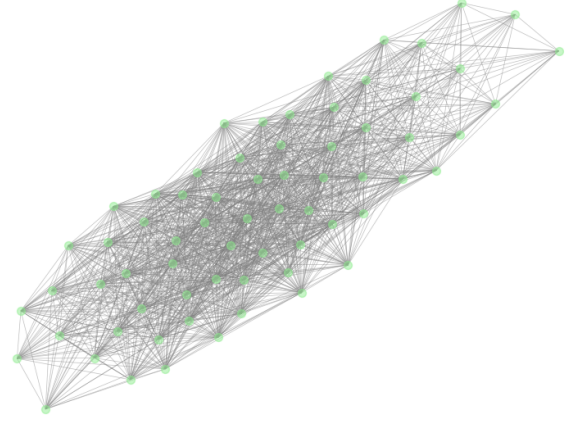


Fig. 2. Florida county graph **after** thresholding ($\lambda = 0.2$) with uniform edge thickness. 1391 edges in total.

Model	batch_size	n_layers	hidden_channels	lr
FF	32	2	32	0.0001
GCN	32	3	32	0.01
GAT	32	3	32	0.01

TABLE I
MODEL HYPERPARAMETERS

D. Training Details

The hyper parameters used for the different models are summarized in Table I. Parameters are optimized using the Adaptive Moment Estimation (Adam) optimizer minimizing the Mean-Square-Error (MSE) loss function. Models are trained using a standard mini-batch stochastic gradient decent method.

VI. RESULTS

A. Training and Validation Loss

Training and validation loss across epochs are summarized in Figures 3 to 5. All models were trained for 500 epochs.

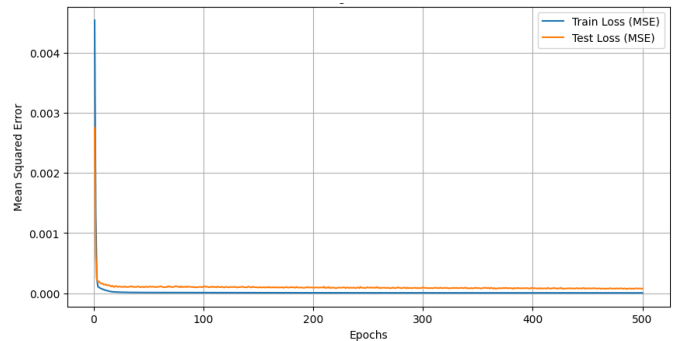


Fig. 3. Train and Test Loss for the FF network model

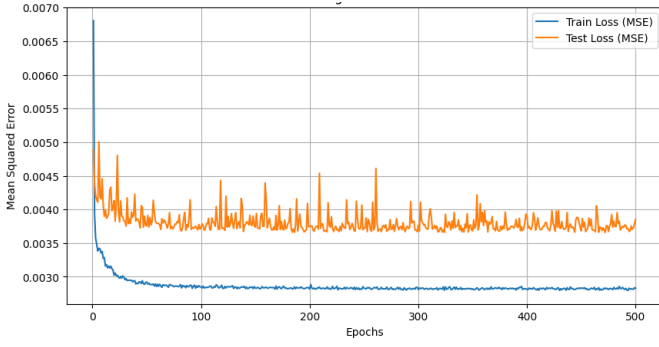


Fig. 4. Train and Test Loss for the GCN model

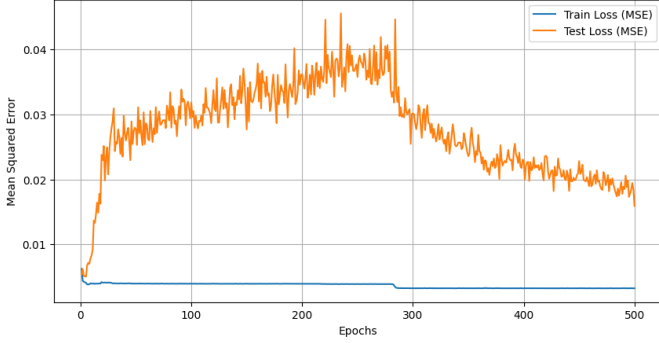


Fig. 5. Train and Test Loss for the GAT model

B. Forecast Accuracy

We report the MAPE and RMSE of the models in Table II. These metrics give insights on how different model architectures influence accuracy in forecasting tasks. First, the baseline feed forward network obtained an adequate MAPE of 1.94% and a RMSE of 665.68 MW. The metrics of the baseline FF-NN model help us determine whether the additional effort of implement more complex graph-based strategies is worthwhile.

The GCN model achieved the best MAPE and RMSE (1.27% and 482.82 MW, respectively). This suggests that the modeling of spatial structures of the electrical network as graph embeddings results in an improvement of about 33% in accuracy for prediction and regression tasks in electricity demand forecasting.

Finally, the GAT model achieved the worst results compared to the baseline with a relatively high 5.09% MAPE and 1529.89 MW RMSE. This suggests that using complex graph models has diminishing returns and may negatively impact accuracy on simple training workflows without specialized hyperparameter tuning strategies.

C. Visual Forecast Comparison

In addition to training loss plots in Figures 3 to 4, a comparison between the predicted and ground-truth values for the first one hundred hours of the testing set time series is shown in Figures 6 to Figure 8.

Model MAPE (%) RMSE (MW)

FF	1.94	665.68
GCN	1.27	482.82
GAT	5.09	1529.89

TABLE II
PERFORMANCE PER MODEL

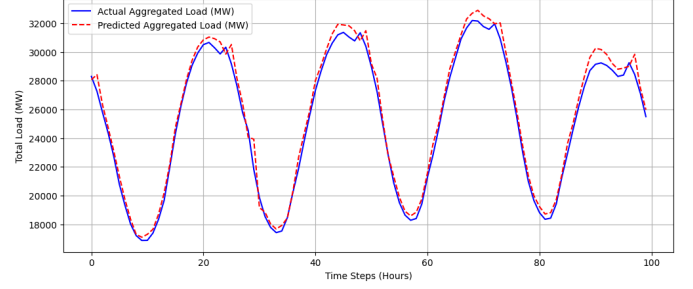


Fig. 6. FF model prediction Forecast vs Actual (First 100 test points)

D. Discussion

Our findings of graph models outperforming linear feed forward networks is consistent with the findings of Campagne *et al.* [1]. However, our results vary when we consider which graph model delivered the best results. While the previous paper found that GCN models perform slightly worse than the baseline and GAT perform slightly better, our results were the opposite: the GCN model outperformed both the baseline and GAT models, with GAT underperforming the baseline.

The causes for the discrepancy might vary from limited training or hyperparameter tuning setup to poor modeling of the GAT network. Visualizations from Figure 8 and the increasing validation loss during training for the GAT network shown in Figure 5 suggest that the GAT model tends to overfit. This result is coherent with that observation that “*in medium-scale settings with stable aggregated targets, architectural simplicity and strong inductive biases can match or even surpass the performance of more sophisticated designs.*” [1]

VII. CONCLUSION

We asked whether modeling Florida counties as a connected graph improves hour-ahead electricity load forecasting, and the answer was yes: the GCN model outperformed the non-spatial FF-NN baseline. And, while dissimilar to other studies,

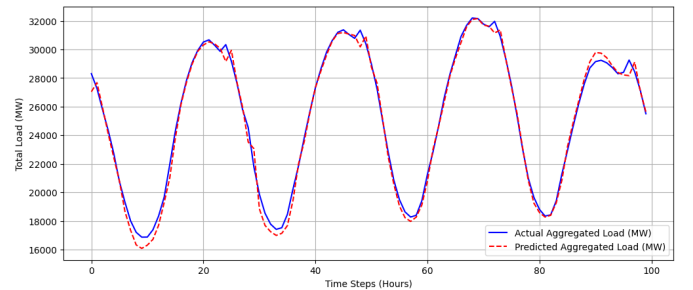


Fig. 7. GCN model prediction Forecast vs Actual (First 100 test points)

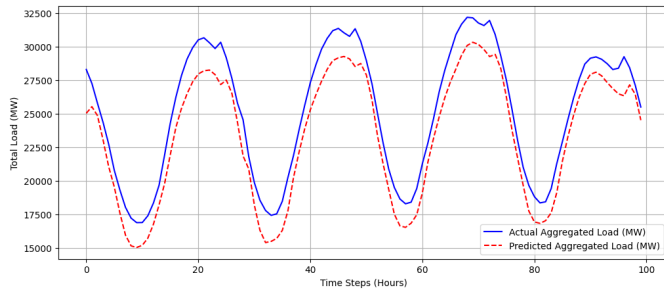


Fig. 8. GAT model prediction Forecast vs Actual (First 100 test points)

we found that the GAT model underperformed, it showed promising results thanks to its attention mechanism that can adapt to changing spatial patterns. More focus is warranted on tuning hyper-parameters of GAT networks to leverage its complex but potentially better interpretability in electricity demand forecasting scenarios.

Future directions we would like to explore include adding weather covariates such as temperature and humidity as node features, testing longer prediction horizons, and analyzing how the GAT attention weights change across seasons to extract interpretable insights about which county relationships drive Florida's electricity demand.

ACKNOWLEDGMENT

We thank the instructors and teaching assistants of EEL 6878 at the University of Central Florida for their guidance throughout this project. Load data are provided by the National Renewable Energy Laboratory through the Open Energy Data Initiative (OEDI).

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